

Lucas A. Lebron

lucaslbron@proton.me | [linkedin.com/in/lucaslebron](https://www.linkedin.com/in/lucaslebron) | github.com/Lucaslbron | [Portfolio](#) | 678-769-8322

EDUCATION

Bachelor of Science in Aerospace Engineering and Mechanical Engineering, Dual Major; Minor in Mathematics
Kennesaw State University, Atlanta, GA | GPA: 3.65 | June 2025 – Fall 2027

Associate's in Science & Technology

University of North Georgia, Gainesville, GA | August 2022 – May 2025

Relevant Coursework: Mathematical Modeling, Rocket Propulsion, Orbital Mechanics, CAE Design, Aircraft Structural Analysis

WORK EXPERIENCE

Undergraduate Research Assistant | Kennesaw State University **June 2026 - Present**

Present Project: Optimization of Magnesium Powder Nanocomposites for Additive Manufacturing Purposes

- **Evaluating** rapid hot compaction workflows against conventional long-duration argon sintering to reduce fabrication time while ensuring mechanical and structural integrity.
- **Characterizing** powder flowability, tap/apparent density, and angle of repose using ASTM standardized Hall Flowmeter apparatuses to validate the feedstock's viability for powder bed fusion additive manufacturing.
- **Conducting** comprehensive material analysis, including Vickers microhardness, Archimedes density, and electrochemical corrosion testing; to optimize the balance between material strength and degradation behavior.

Undergraduate Research Assistant | Kennesaw State University

Nov 2025 - Present

Present Project: Optimization of PEO Coatings on Magnesium ZK60 Alloy

- **Conducting** experimentation on Plasma Electrolytic Oxidation (PEO) variables to improve corrosion protection and electrical/thermal insulation properties of aerospace-grade alloys.
- **Developing** Standard Operating Procedures (SOPs) for coating application and subsequent material testing.
- **Supervising** and mentoring a student assistant in laboratory safety, sample preparation, and data documentation protocols.

AMS Pools | Pool Manager

Mar 2022 - Sept 2024

- **Managed** operations across 20+ pool decks, overseeing 10–15 employees daily while ensuring patron and HOA satisfaction.
- **Maintained** pool systems, chemical balance, safety compliance, and conflict resolution; trained 150+ staff annually.

PROJECTS

C++ Aerodynamic Panel Solver

June 2026 -Present

- **Developing** an object-oriented C++ architecture utilizing custom classes to represent aerodynamic panels, nodes, and boundary conditions for arbitrary 2D airfoil geometries.
- **Engineering** a custom linear algebra solver utilizing LU decomposition (Gaussian elimination) to efficiently solve the dense system of equations generated by the panel influence matrix.
- **Implementing** robust File I/O protocols to parse standard Selig/UIUC airfoil coordinate .dat files and export computed pressure coefficient distributions to CSV for post-processing and visualization.

Subscale Liquid Rocket Booster | Rocketry Club

Jan 2026 - Present

Current Project: Impinging Injector Atomization Optimization

- **Designing** and optimizing an Impinging Injector to enhance atomization and mixing efficiency in the combustion chamber.
- **Developing** the overall structural architecture of the liquid booster engine, ensuring mechanical integrity under high pressure and high temperature operating conditions.
- **Integrating** Design for Manufacturing (DFM) principles to streamline the transition from initial CAD models to physical prototype fabrication.
- **Utilizing** advanced modeling tools like Solidworks to create detailed assemblies and technical drawings for the injector manifold, and ANSYS to model fluid ejection angles and convergence points.

Electric Airbike | CAD Design & Multidisciplinary Analysis

June 2026 - August 2026

- **Designed** an Aluminum 6061-T6 electric airbike chassis, optimizing structural geometry for operator ergonomics, component packaging, and wheel clearances.
- **Performed** FEA, CFD, and fatigue analysis to evaluate structural and aerodynamic performance; reduced chassis maximum stress from 52,330 psi to 2,786 psi and deflection from 5.512 in to 0.0344 in.
- **Conducted** thermal and fluid-system analyses of the battery pack and heat exchanger, modeling heat generation, phase-change cooling, and fluid flow requirements for battery thermal management.

SKILLS

CAD & Simulation: SolidWorks (CFD), AutoCAD, Autodesk Inventor, ANSYS (FEA), NASTRAN

Programming: C++, MATLAB, Python, Proxmox

Laboratory & Materials: PEO Coatings, ASTM & ISO Standardized Testing, Electrochemical Corrosion Testing, Scanning electron microscopy/EDS, X-Ray Diffraction Analysis, Potentiodynamic Polarization, Electrochemical Impedance Spectroscopy

Manufacturing & Systems: Design for Manufacturing (DFM), Systems Integration, Prototype Fabrication, Technical Documentation, Bill of Materials (BOM), DOE documentation